

Xilin Mao

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Education

School of Mathematical Sciences, University of Chinese Academy of Sciences, BS in Mathematics	Sept 2020 – Jul 2024
• GPA: 3.87/4.0 • Coursework: Complex Geometry, Algebraic Topology	
Qiuzhen College, Tsinghua University, Ph.D in statistics	Sept 2024 – Jun 2029(expected)
• Advisor: Prof. Yuhong Yang • Research area: Machine Learning, Causal Inference and Model Selection • GPA: 3.8/4.0(until now)	

Projects

Model Averaging in Policy Learning	May 2026 – Now
• Develop model averaging methods under some criterions in the field of policy learning with or without unconfoundedness assumption.	
Data Combination in Causal Inference (Submitted to the Journal of the American Statistical Association (JASA))	May 2025 – Aug 2026
• Develop efficient estimators for the difference of CATEs between an Observational Study and an Randomized Trail, with unobserved confounders appearing in the Observational Study. • Tools Used: semiparametric inference, M-estimators, empirical processes, sensitivity analysis.	
Alpha factor mining with LLMs	Apr 2026 - Jun 2026
• I surveyed this field and reproduced two papers. • Based on the results of the reproduction, I propose a new algorithm.	
Homotopy Theory in Complex Geometry	Nov 2023 – May 2024
• My undergraduate dissertation, supervised by Pro. Songyan Xie. • A survey about model category structure and Oka theory.	

Awards and Honors

Team Competition Bronze Medal of S.-T.Yau College Student Mathematics Contest, Yau Mathematical Sciences Center, Tsinghua University	Jun 2023
First-Class Academic Scholarship, University of Chinese Academy of Sciences	Nov 2022
Second-Class Academic Scholarship, University of Chinese Academy of Sciences	Nov 2023

Technologies

Programming Languages: Good at Python and R, know something about C, Matlab.